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Correcting HVAC Effect on Pressure Sensor Measurement

Abstract

A location context indicates whether a computing device is in an HVAC environment or a non-HVAC environment. When the computing device is in the HVAC environment, an expected altitude of the computing device is obtained, and a measured altitude of the computing device is determined. An altitude difference is determined between the expected altitude and the measured altitude. The HVAC effect is determined based on the altitude difference. A pressure sensor of the computing device is calibrated based on the HVAC effect, or a corrected measured altitude of the computing device is estimated based on the HVAC effect. In some cases, instead of the altitude difference, the HVAC effect is determined based on a pressure difference.

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Background/Summary

RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Patent Application No. 63/558,255, filed Feb. 27, 2024, all of which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] Determining the exact 3D location of a device (e.g., typically a mobile device, such as a smartphone or other computing device that is carried and operated by a user) in an environment can be quite challenging, especially when the computing device is located in an urban environment or is located within a building or a moving vehicle. Imprecise estimates of the computing device's altitude, for example, may have life-or-death consequences for the user of the computing device since the imprecise altitude estimate can delay emergency personnel response times as they search for the user on multiple floors of a building. In less dire situations, imprecise altitude estimates can lead a user to the wrong area in an environment.

[0003] Environments affected by artificial pressurization, pressure turbulence, and other pressure variations can hinder barometric-based altitude determination. Such environments generally include vehicles and buildings that have HVAC (heating, ventilation, and air-conditioning) capabilities or facilities, though not all such enclosed environments with HVAC necessarily demonstrate significant adverse pressure behavior at all times (e.g., depending on whether vents are closed or open, or air is being actively circulated, among other possible situations). Since barometric-based altitude solutions require an accurate measurement of ambient pressure by a pressure sensor of the device, it is important to flag the data for such environments to alert that the measurement may be affected by HVAC, if detected, so that applications will not use the affected data.

SUMMARY

[0004] In some embodiments, a system or method involves determining a location of a computing device; determining a location context for the computing device, wherein the location context indicates whether the computing device is in an HVAC (heating, ventilating, or air-conditioning) environment that can be subject to an HVAC effect or is in a non-HVAC environment that is not subject to the HVAC effect; upon determining that the computing device is in the non-HVAC environment, not determining the HVAC effect; and upon determining that the computing device is in the HVAC environment: obtaining an expected altitude of the computing device based on the location of the computing device; determining a measured altitude of the computing device based on a reference pressure and a measured pressure from a pressure sensor of the computing device; determining an altitude difference between the expected altitude and the measured altitude; determining the HVAC effect based on the altitude difference; and calibrating the pressure sensor based on the HVAC effect or estimating a corrected measured altitude of the computing device based on the HVAC effect. In some embodiments, a pressure difference is used instead of an altitude difference.

[0005] In some embodiments, the determining of the location of the computing device to the determining of the altitude difference is repeated for each time point of a plurality of time points; and the determining of the HVAC effect further includes: for each time point of the plurality of time points, mapping the altitude difference with respect to the location context to generate mapped altitude differences; determining a first cluster of the mapped altitude differences for when the

computing device is in the HVAC environment; determining a second cluster of the mapped altitude differences for when the computing device is in the non-HVAC environment; and determining the HVAC effect based on the first cluster of mapped altitude differences and the second cluster of mapped altitude differences. In some embodiments, for each time point of the plurality of time points, the altitude difference is mapped with respect to the time point and the location context to generate the mapped altitude differences. In some embodiments, a magnitude of the HVAC effect is determined that is a difference between a first average of the first cluster of mapped altitude differences and a second average of the second cluster of mapped altitude differences.

[0006] In some embodiments, the location context is determined to be in the HVAC environment which is a building. In some embodiments, determining the location context for the computing device further includes: determining that the computing device is in the building by mapping the location of the computing device to a building database, which indicates that the computing device is within the building; determining that the computing device is in the building by determining that the computing device is wirelessly connected to a wireless access point that is known to be in the building, which indicates that the computing device is within the building; or determining that the computing device is in the building by determining that a strength of a satellite signal is attenuated from a level that it should be if the computing device were outdoors, which indicates that the computing device is within the building. In some embodiments, determining the location context for the computing device further includes: determining that the location context is in the HVAC environment which is a vehicle. In some embodiments, determining the location context for the computing device further includes: determining that the computing device is in the vehicle by mapping the location of the computing device to a road database, which indicates that the computing device is on a road and within the vehicle; determining that the computing device is in the vehicle by determining that the computing device is communicatively connected to the vehicle, which indicates that the computing device is within the vehicle; determining that the computing device is in the vehicle by determining that the computing device is moving faster than a walking, running or biking speed, which indicates that the computing device is within the vehicle; determining that the computing device is in the vehicle by determining that the computing device is changing cell tower connections, which indicates that the computing device is within the vehicle; determining that the computing device is in the vehicle by obtaining an activity context from a sensor hub of the computing device that indicates that the activity context is driving, which indicates that the computing device is within the vehicle; or determining that the computing device is in the vehicle by determining that the computing device is in a parking lot, which indicates that the computing device is within the vehicle.

[0007] In some embodiments, determining the location context for the computing device further includes: determining a device temperature of the computing device; obtaining a reference temperature based on the location of the computing device; and determining the location context by comparing the device temperature and the reference temperature. In some embodiments, determining the location context for the computing device further includes: determining that the reference temperature and the device temperature are within a threshold temperature range from each other; determining that the reference temperature and the device temperature are both less than a cold threshold temperature; and determining that the computing device is in the non-HVAC environment. In some embodiments, determining the location context for the computing device further includes: determining that the reference temperature and the device temperature are within a threshold temperature range from each other; determining that the reference temperature and the device temperature are both greater than a hot threshold temperature; and determining that the computing device is in the non-HVAC environment. In some embodiments, the location context further indicates whether the computing device is in the HVAC environment, the non-HVAC environment, or an uncertain environment; and determining the location context for the computing

device further includes: determining that the reference temperature and the device temperature are within a threshold temperature range from each other; determining that the reference temperature and the device temperature are both greater than a cold threshold temperature and less than a hot threshold temperature; and determining that the computing device is in the uncertain environment. In some embodiments, determining the location context for the computing device further includes: determining that the reference temperature is warmer than the device temperature; determining that the device temperature is greater than a cold threshold temperature and less than a hot threshold temperature; and determining that the computing device is in the HVAC environment. In some embodiments, determining the location context for the computing device further includes: determining that the reference temperature is colder than the device temperature; determining that the device temperature is greater than a cold threshold temperature and less than a hot threshold temperature; and determining that the computing device is in the HVAC environment. In some embodiments, the location context further indicates whether the computing device is in the HVAC environment, the non-HVAC environment, or an uncertain environment; and determining the location context for the computing device further includes: determining that the reference temperature is colder than the device temperature; determining that the device temperature is near a cold threshold temperature within a low-end temperature deviation amount; and determining that the computing device is in the HVAC environment or the uncertain environment. In some embodiments, the location context further indicates whether the computing device is in the HVAC environment, the non-HVAC environment, or an uncertain environment; and determining the location context for the computing device further includes: determining that the reference temperature is warmer than the device temperature; determining that the device temperature is near a hot threshold temperature within a high-end temperature deviation amount; and determining that the computing device is in the HVAC environment or the uncertain environment. In some embodiments, determining the location context for the computing device further includes: determining that the reference temperature is colder than the device temperature; determining that the device temperature is below a cold threshold temperature; and determining that the computing device is in the non-HVAC environment. In some embodiments, the location context further indicates whether the computing device is in the HVAC environment, the non-HVAC environment, or an uncertain environment; and determining the location context for the computing device further includes: determining that the reference temperature is warmer than the device temperature; determining that the device temperature is above a hot threshold temperature; and determining that the computing device is in the uncertain non-HVAC environment. In some embodiments, determining the location context for the computing device further includes: determining the location context by comparing the device temperature and the reference temperature and further determining how the device temperature changes over time.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a simplified environment in which a barometric pressure sensor within a computing device can be calibrated or used to determine an altitude of the computing device, some of which may be subject to HVAC effects, in accordance with some embodiments.

[0009] FIG. 2 is a simplified flowchart of a procedure for correcting measurements of a barometric pressure sensor (of the computing device shown in FIG. 1) based on HVAC effects, in accordance with some embodiments.

[0010] FIG. 3 is a simplified flowchart of a procedure for determining the presence or magnitude of HVAC effects for use in the procedure shown in FIG. 2, in accordance with some embodiments.

[0011] FIG. 4 is a simplified graph of mapped altitude differences vs. time of day for different

location contexts illustrating data clusters for use in the procedure shown in FIG. 3, in accordance with some embodiments.

[0012] FIGS. 5-7 are simplified flowcharts of example procedures for determining a likely location context of the computing device for use in the procedure shown in FIG. 2, in accordance with some embodiments.

[0013] FIG. 8 is a simplified graph of a reference temperature vs. a device temperature illustrating location contexts and changes in location contexts for use in the procedure shown in FIG. 7, in accordance with some embodiments.

[0014] FIG. 9 shows simplified graphs of a reference temperature and a device temperature vs. time of day illustrating relative differences in the reference temperature and the device temperature depending on location contexts and changes in location contexts for interpreting the data shown in FIG. 8, in accordance with some embodiments.

[0015] FIG. 10 shows simplified schematic diagrams of a transmitter, a computing device, and a server for use in the environment shown in FIG. 1, in accordance with some embodiments.

DETAILED DESCRIPTION

[0016] Conventional systems or techniques for calibrating a barometric/atmospheric pressure sensor in a computing device or determining an altitude of the computing device when HVAC (heating, ventilation, and air-conditioning) effects may be a factor generally use data that is densely available in time, i.e., a data rich scenario in which data is collected relatively regularly and frequently. An example of such a technique is described in U.S. Pat. No. 10,477,358, which is incorporated herein by reference as if fully set forth herein. Such conventional techniques typically detect a sudden change in barometric pressure, which may indicate that the computing device has likely moved from a non-HVAC environment (i.e., an environment that is not subject to HVAC effects or biases) to an HVAC environment (i.e., an environment that is subject to HVAC effects) or vice versa or from a low HVAC environment (i.e., an environment that is subject to a relatively low HVAC effect) to a high HVAC environment (i.e., an environment that is subject to a relatively high HVAC effect) or vice versa. The “suddenness” of the pressure change is a consideration, because the transition between the two environments typically takes only a few seconds, e.g., the time it typically takes for a person to pass through a door. Thus, the sudden pressure change can generally be detected only with data that is collected relatively regularly and frequently, so that the point at which the pressure change occurs (i.e., when the change between the non-HVAC environment and the HVAC environment occurs) is evident in the pressure data with respect to time when the change is not accompanied by an apparent corresponding sudden altitude change. Thus, when the sudden pressure change is detected as a result of a movement of the computing device from one environment to another, it is readily apparent which of the two environments was subject to the HVAC effects, and any calibration or altitude determination technique can then properly take the HVAC effect into account for the relevant environment.

[0017] However, many situations can occur when the available data is relatively sporadic, instead of regularly occurring, thereby rendering such conventional systems or techniques inadequate, inappropriate or insufficient for producing a reliable calibration or altitude result. Sporadic data is data that is generally collected with insufficient regularity (i.e., not collected at regular intervals) and/or frequency (i.e., the data is too sparse or spread out with respect to time or distance) to render the sudden pressure change clearly evident in the pressure data. Without such regularity and frequency, the pressure change with respect to time might not indicate that the change was “sudden”, as typically happens with a transition from one environment to another. For example, regular/frequent data that tracks a computing device entering a single-story building and presenting a sharp jump in barometric pressure could indicate that the building is subject to HVAC effects, but if there were only one data collection event (i.e., an example of sporadic data) inside the building with no other context, there would not be enough information to draw a conclusion of whether there is an HVAC effect, since a pressure delta would require at least two measurements. In another

example, if the time period between successive data collections is so large (i.e., another example of sporadic data) that a ratio of the measured pressure change with respect to time is negligible, there would also not be enough information to draw a conclusion of whether there is an HVAC effect. There is, thus, a temporal dimension or a spatial dimension or both in the qualitative and quantitative difference between sporadic data and regular data. Sporadic data is less dense or less consistent than regular or dense data. Sporadic data is not collected at regular intervals (i.e., is randomly collected) or is more spread out in time or distance than regular data. The techniques described herein have the advantage or improvement of being able to work with both sporadic data and regular data, instead of just regular data.

[0018] In a simplified summary of the techniques described herein, a likely location context for the computing device (and the user thereof) is first determined. In other words, it is determined whether the computing device is apparently in a vehicle (whether moving or stopped), a building, or some other environment that is potentially subject to HVAC effects. Then a measured altitude or pressure at such times is compared to those at other times when the computing device is apparently NOT in a vehicle, a building, or some other environment that is potentially subject to HVAC effects. In this manner, magnitudes of altitude differences or pressure differences can be compared across different data collection environments. With the altitude differences or pressure differences, the existence of HVAC push/pull effects to the ambient pressure are identified and/or quantified. Identified instances where there exists a push or pull in the pressure measurement of the barometric pressure sensor of the computing device aid in calibration of the pressure sensor or accurate determination of an altitude of the computing device. (A “push” is defined herein to be an increase in the measured pressure, which corresponds to a lower calculated altitude compared to the actual altitude; and a “pull” is defined herein to be a decrease in the measured pressure, which corresponds to a higher calculated altitude compared to the actual altitude.)

[0019] Since the techniques described herein can work with both sporadic data and regular data, this improves upon the opportunities for and accuracy of altitude estimations of the computing device. Additionally, some aspects of the disclosure herein improve the determination of a more appropriate uncertainty in the estimated altitude of the computing device. In addition, calibration of the pressure sensor is improved by helping to filter out unreliable or less desirable data points or ensuring that the uncertainty in the calibration result is more appropriate. The resulting improved calibration value for the pressure sensor of the computing device improves the accuracy of the estimated altitude of the computing device. Therefore, the disclosure herein represents a specific set of rules in a process flow that provides significant advantages in the fields of calibration and position/altitude determination.

[0020] The benefits or advantages of the present disclosure are particularly significant in a situation where a user calls an emergency service such as **911** from within a high-rise office building during a life-or-death situation. In such a situation, a floor designator that indicates which floor of the building the user is on may be significantly more useful to the first responders or emergency personnel trying to reach the user as compared to receiving a raw estimated altitude value reported by the user's computing device. Therefore, the improved technique described herein for determining the altitude of the computing device (e.g., with better calibration of the pressure sensor therein or improved altitude estimation) can be combined with a technique for determining the floor most likely to correspond to the determined altitude (e.g., as described in US Patent Publication No. 2023/0384087), and thereby improve emergency service response times. (US Patent Publication No. 2023/0384087 is incorporated herein by reference as if fully set forth herein.) In this case, if the user is calling from within a multi-floor building, then the emergency 911 service would require the floor number in order to properly dispatch an emergency medical team to the correct floor. However, if the user cannot provide the floor number, then the 911 app on the computing device or a server must provide it, i.e., correct knowledge of the floor number (or altitude of the computing device) can be critical for the emergency medical team to search for and

find the user on the correct floor in a timely manner. Since the building is likely to be an environment that is subject to HVAC effects, the present disclosure can aid in making the determination of the most likely floor or a more accurate range of floors.

[0021] Additionally, there are a wide range of useful software applications that operate on computing devices and that require knowledge of the altitude of the computing device to operate properly or effectively. In some examples, some such applications simply inform the user of the computing device of their current location, including their altitude, for whatever personal purpose the user might have or feel a need for. In a specific example, the user might want to use an app on the computing device to receive electronic discounts or special advertisements for a retail store upon entering the store. However, if the store is in a shopping mall that has more than one floor, then the app needs to know which floor the computing device is on in order to operate properly, i.e., the app requires knowledge of the altitude of the computing device so that it can compare this altitude with the altitudes of the floors and make a proper determination of whether the user is likely to be in the correct store. Since the shopping mall or the store could be an environment that is subject to HVAC effects, the present disclosure can aid in making the determination of whether the user and computing device are in the correct store.

[0022] The examples in the previous paragraphs highlight the importance, even life-saving criticality, of a good altitude determination for a computing device in various situations, especially in situations that are subject to HVAC effects. However, there is usually some amount of uncertainty in the determined altitude value. Given the importance of a proper altitude determination, it is necessarily of equal importance to provide the altitude value with a high degree of certainty (i.e., a low degree of uncertainty) or a good determination of the uncertainty. In the critical example given above, the emergency medical team can be provided with the likely floor number for the computing device and the user, but with a further level of likelihood of whether the user could be on the next floor above or below, so that the emergency medical team can know whether to expand their search for the user to these other floors. Proper consideration of potential HVAC effects, even with sporadic data as described herein, can significantly improve the search for the user.

[0023] FIG. 1 shows a simplified example operational environment **100** in which a barometric pressure sensor within a computing device **102** (e.g., a mobile user device) can be calibrated or used to determine an altitude of the computing device **102**, in accordance with some embodiments. The operational environment **100** includes at least one of the computing devices **102**, one or more system servers **104**, and a network of terrestrial transmitters or reference pressure stations **106**. The system server **104** exchanges communications with various devices, such as the computing device **102**. Each of the reference pressure stations **106** and the computing device **102** may be located at different altitudes or depths that are inside or outside various natural or manmade structures (e.g., buildings **108** and **110**), relative to different elevations throughout a terrain **112**, as illustrated by examples in FIG. 1. Positioning signals **114** and **116** are transmitted from the reference pressure stations **106** and satellites **118** (e.g., part of a Global Navigation Satellite System (GNSS)), respectively, and are subsequently received by the computing device **102** using known transmission technologies. For example, the reference pressure stations **106** may transmit the signals **114** using one or more common multiplexing parameters that utilize time slots, pseudorandom sequences, frequency offsets, or other approaches, as is known in the art or otherwise disclosed herein. Examples of possible hardware, software and data components in the reference pressure stations **106**, the computing device **102**, and the system server **104** are shown in FIG. 10, as described below. In particular, each reference pressure station **106** and computing device **102** may include atmospheric sensors (e.g., barometric pressure sensors and temperature sensors, as appropriate) for generating measurements of atmospheric conditions (e.g., atmospheric pressure and temperature) that are used to estimate an altitude of the computing device **102** or to calibrate the barometric pressure sensor therein.

[0024] As a person/user **120** carrying the computing device **102** moves throughout such an example operational environment **100**, the computing device **102** may experience a variety of activities or diversity of locations both indoors and outdoors. For example, the computing device **102** may be moving rapidly (e.g., location changes significantly in a relatively short time) when it is in a vehicle (as on a road at **122**) that the user is driving or riding in. In another example, the computing device **102** may be still or moving slowly (e.g., as indicated by an accelerometer or GNSS signals or other positioning sensor within the computing device **102**) when the user **120** is simply standing, sitting, or walking (e.g., as in a building at **124** or **126**, in a parked vehicle at **128**, or walking outdoors at **130** or **132**).

[0025] An environment characterized by HVAC or which is subject to HVAC effects is an environment where the air is warmed or cooled and/or filtered or cycled, usually by a specialized HVAC system. HVAC systems can add or remove additional air mass into or from an enclosed environment in a controlled fashion and can cause such an environment to increase or decrease in pressure—a well-understood phenomenon. However, artificial pressurization (positive or negative) can strongly affect a barometric-based altitude system, where barometric pressure measured by a barometric pressure sensor of a computing device is used in conjunction with ambient information such as a reference pressure and temperature to calculate the altitude of the computing device. Altitudes measured in such environments can typically be biased lower or higher due to the HVAC push or pull, respectively, so the true altitude of the computing device is not directly reflected by the raw pressure data from the pressure sensor thereof.

[0026] In FIG. **1**, the environments that are potentially subject to HVAC effects are the buildings **108** and **110** (having HVAC systems **109** and **111**, respectively) and the vehicles at **122** or **128**. However, other types of environments not represented in FIG. **1** may also be subject to HVAC effects, e.g., any enclosed environment that has an HVAC system. Additionally, a further complication for the vehicles may be caused by high wind speed effects due to the movement of the vehicles.

[0027] FIG. **2** is a simplified flowchart of an example process **200** for correcting measurements of a barometric pressure sensor (of the computing device **102**) based on HVAC effects, in accordance with some embodiments. The process **200** can work with sporadic data as well as regular data, so the process **200** enables or embodies all of the advantages, benefits or improvements described herein. The particular steps, combination of steps, and order of the steps for this process are provided for illustrative purposes only. Other processes with different steps, combinations of steps, or orders of steps can also be used to achieve the same or similar result. Features or functions described for one of the steps performed by one of the components may be enabled in a different step or component in some embodiments. Additionally, some steps may be performed before, after or overlapping other steps, in spite of the illustrated order of the steps.

[0028] Some of the steps of the process **200** use the location of the computing device **102**. Therefore, in some embodiments, upon starting or at any appropriate point, the process **200** determines (at **202**) the 2D or an approximate 3D location of the computing device **102** using conventional location/position techniques.

[0029] At **204**, the process **200** determines the likely location context of the computing device **102**. The location context can be determined using a variety of different techniques. FIGS. **5-9**, described below, illustrate three example techniques for determining the location context. The processes of FIGS. **5** and **6**, for example, use the location determined at **202** to determine the location context. In particular, the process of FIG. **5** determines whether the computing device **102** is in a building, and the process of FIG. **6** determines whether the computing device **102** is in a vehicle. On the other hand, FIGS. **7-9** illustrate a process that does not use a location context that is based on the location determined at **202** but infers the location context based on temperature considerations, so this process does not specify the type of location context, e.g., in a building or vehicle. In some embodiments, therefore, at **204**, the process **200** performs any one or more of the

processes of FIGS. 5-9.

[0030] The determined location context is generally related to whether the computing device **102** is likely to be in an environment that can be subject to an HVAC effect, e.g., inside a building, a vehicle, or other enclosed environment that might have an HVAC system. In some embodiments, therefore, the returned result of any one or more of the processes of FIGS. 5-9 may state that the computing device **102** either is or is not in a building (FIG. 5), a vehicle (FIG. 6), or any other environment that can be subject to an HVAC effect (FIGS. 7-9).

[0031] At **206**, the process **200** determines whether the location context of the computing device **102** is an environment that can be subject to an HVAC effect, e.g., as indicated by the returned result of any one or more of the processes of FIGS. 5-9 at **204**. Therefore, the determination at **206** is negative or “No” if the returned result of the process of FIG. 5 states that the computing device **102** is not in a building, the returned result of the process of FIG. 6 states that the computing device **102** is not in a vehicle, and the returned result of the process of FIGS. 7-9 states that the computing device **102** is not in any environment that can be subject to an HVAC effect. In this case, the process **200** proceeds at **218** to perform any functions that do not take into account HVAC effects, e.g., calibrate the pressure sensor of the computing device **102** or estimating a measured altitude of the computing device **102** without any adjustment for HVAC effects.

[0032] On the other hand, the determination at **206** is positive or “Yes” if the returned result of the process of FIG. 5 states that the computing device **102** is in a building, the returned result of the process of FIG. 6 states that the computing device **102** is in a vehicle, or the returned result of the process of FIGS. 7-9 states that the computing device **102** is in any environment that can be subject to an HVAC effect. In this case, the process **200** proceeds to perform the following functions at **208-216** to take into account or quantify (i.e., determined the magnitude of) potential HVAC effects.

[0033] In some embodiments, at **208**, the process **200** obtains an expected altitude (i.e., a lookup altitude) of the computing device **102**. The expected altitude can be determined by a conventional technique based on the location of the computing device **102** (e.g., as determined at **202**) and an altitude indicated for that location in a terrain or building database (e.g., a terrain lookup, a building floor altitude lookup, etc.). In some embodiments, the expected altitude can be obtained with help from the user of the computing device **102**, e.g., by receiving floor level input from the user when the user is in a multi-level building or shopping mall and then performing a building floor altitude lookup. In some embodiments, at **210**, the process **200** determines a measured altitude (i.e., a barometric altitude) of the computing device **102** using conventional techniques, e.g., using a barometric formula based on a measured pressure value from the pressure sensor of the computing device **102** and a reference pressure and reference temperature obtained from a network of reference atmospheric sensors (e.g., the reference pressure stations **106** in FIG. 1) in the same general area as the location of the computing device **102**. The measured pressure value may or may not be subject to HVAC effect, so if an HVAC effect is determined in the subsequent steps, then the measured altitude (i.e., an initial altitude) calculated from the measured pressure value will be corrected at the end of the process **200** (i.e., a corrected measured altitude).

[0034] In some embodiments, at **212**, the process **200** determines an altitude difference between the expected altitude and the measured altitude. The altitude difference may be calculated as: altitude difference=measured altitude–expected altitude. For example, if the measured altitude was 19 m and the expected altitude based on terrain data at the given location of the computing device **102** was 7 m, then assuming that the computing device **102** is offset above the ground by about 1 m, the altitude difference is $19-(7+1)=11$ m. However, if the expected altitude of the computing device **102** could not be unambiguously determined via a terrain or building data lookup or other available means, such as user intervention (for example, in a situation where the computing device **102** is within a building having multiple floors), then the current dataset using the location context of the computing device **102** might not be useable and may have to be discarded.

[0035] At **214**, the process **200** determines and stores the presence or magnitude of the HVAC effect for a given environment based on the altitude difference. In some embodiments, the process **200** at steps **202-212** is repeated for multiple time points in order to generate multiple datasets for determining the presence of or magnitude of the HVAC effects of the given environment. An example process for determining the presence or magnitude of HVAC effects at **214** using multiple datasets is described below with reference to FIGS. **3** and **4**.

[0036] In some embodiments, a database is generated for the magnitude of the HVAC effect for known buildings or vehicles. Therefore, when any computing device **102** is in a known building or vehicle, then the magnitude of the HVAC effect can be looked up in the database and used to determine the altitude of the computing device **102** and, if necessary, to calibrate the pressure sensor therein. Additionally, the database may store the HVAC effect magnitude relative to the days of the week, the outdoor temperature, the season, or other appropriate parameter.

[0037] At **216**, the process **200** uses the determination made at **214**. In some embodiments, the process **200** uses the magnitude of the HVAC effect to estimate a corrected measured altitude of the computing device **102**. For example, the measured altitude that was determined at **210** is adjusted or corrected by subtracting the magnitude of the HVAC effect therefrom, and an associated uncertainty is also corrected based on an uncertainty due to the HVAC effect. In some embodiments, it may be necessary to calibrate the pressure sensor in spite of the data being affected by HVAC effects, so the process **200** uses the magnitude of the HVAC effect to calculate a calibration value and a calibration uncertainty value for the pressure sensor. For example, a conventionally determined calibration value and calibration uncertainty value are further adjusted for the HVAC effect and the uncertainty that it represents. In some embodiments, the process **200** uses the mere presence of the HVAC effect to determine not to calibrate the pressure sensor or estimate the altitude of the computing device **102** using the HVAC-affected data, e.g., due to the possibility that the HVAC effect would render the resulting calibration value or altitude too uncertain. In some embodiments, if there is a potential HVAC effect present, then it may be preferable to use a previously determined calibration value and uncertainty.

[0038] The description above of steps **208-214** describes an altitude-based version of the process **200**. However, in some embodiments, a pressure-based version may be employed. In this case, the process **200** need not obtain the expected altitude of the computing device **102** at **208**. Therefore, if the determination at **206** is positive or “Yes”, then the process **200** proceeds to **210**.

[0039] At **210**, the process **200** determines at least two measured pressures, which may be from the same computing device **102** or two different computing devices **102**, i.e., a single-device embodiment and a two-device embodiment. (The two-device embodiment may generally be performed if the two devices are known to be well-calibrated.) One of the pressure measurements is taken when the location context of the computing device **102** is an environment that is not subject to an HVAC effect, and the other pressure measurement is taken when the location context of the (same or different) computing device **102** is an environment that can be subject to an HVAC effect. Additionally, the time difference between the two pressure measurements is preferably relatively brief, e.g., about 10 minutes or less, so that any change in the actual ambient atmospheric pressure between the two measurements is likely to be negligible (e.g., about 5 Pa or less). In this situation, it can be assumed that the ambient pressure has not changed significantly in this time period. Furthermore, unless it is known that the altitudes of the two locations are the same, then the distance between the two locations at which the two measurements are taken is preferably relatively small, so that the altitude of the computing device **102** at the two locations can be assumed to be almost the same. Thus, it can be assumed that the user has not traversed with the computing device **102** (for the single-device embodiment) a large enough distance (e.g., moved about 200 m or more) or that the two computing devices **102** are not too far apart (for the two-device embodiment) for a relatively large horizontal pressure gradient (e.g., about 10 Pa/km or more) to be significant. If these conditions are met, then the process **200** can compare the pressure

difference between the two measurements instead of the altitude difference, and the pressure difference can be assumed to be primarily caused by the difference between the HVAC environment and the non-HVAC environment.

[0040] Therefore, at **212**, the pressure difference is determined or calculated as: pressure difference=first barometric pressure–second barometric pressure. As an example, if the location context of the computing device **102** at a first point is determined to be in a parked vehicle and the measured pressure is 100100 Pa, and the location context of the (same or different) computing device **102** at a second point (less than 200 m away from and less than 10 min after the first point) is determined not to be in the parked vehicle (or a building or other HVAC environment) and the measured pressure is 100050 Pa, then the pressure difference is 50 Pa.

[0041] An additional benefit of the pressure-based version of the process **200** is that a reference pressure and temperature lookup is not required as it is for determining the measured altitude described above.

[0042] FIG. **3** is a simplified flowchart of a process **300** (both an altitude-based version and a pressure-based version) for determining the presence or magnitude of HVAC effects at step **214** (FIG. **2**), in accordance with some embodiments. The particular steps, combination of steps, and order of the steps for this process are provided for illustrative purposes only. Other processes with different steps, combinations of steps, or orders of steps can also be used to achieve the same or similar result. Features or functions described for one of the steps performed by one of the components may be enabled in a different step or component in some embodiments. Additionally, some steps may be performed before, after or overlapping other steps, in spite of the illustrated order of the steps.

[0043] In some embodiments of the altitude-based version, at **302**, the process **300** associates the altitude differences (e.g., as determined at **212** for multiple time points) with the corresponding time points and location contexts. An example for how the altitude differences, time points, and location contexts can be associated is shown in FIG. **4**. FIG. **4** is a simplified graph **400** of simulated data for mapped or graphed altitude differences vs. time of day for different location contexts that shows clusters of the data.

[0044] The data shown with circles represent a location context of the computing device **102** being in a building at the same known floor. The data shown with triangles represent a location context of the computing device **102** being in a vehicle that is travelling relatively fast. The data shown with exes (“X”) represents a location context of the computing device **102** being in a vehicle that is still or travelling relatively slowly. The data shown with squares represents a location context of the computing device **102** not being in either a building or a vehicle.

[0045] The data is generally distributed in three data clusters **402**, **404** and **406**, which can be determined using conventional data clustering techniques. For example, at **304**, the process **300** determines a first cluster (e.g., a middle data cluster **402**) of the mapped altitude differences for when the computing device **102** appears not to be in an environment that can be subject to an HVAC Effect. Additionally, at **306**, the process **300** determines one or more second clusters (e.g., data clusters **404** and **406**) of the mapped altitude differences for when the computing device **102** is in an environment that can be subject to an HVAC Effect. The data cluster **404** is a high cluster that represents a potential HVAC pull effect, since an HVAC pull results in a lower air pressure, which is generally associated with a higher altitude; and the data cluster **406** is a low cluster that represents a potential HVAC push effect, since an HVAC push results in a higher air pressure, which is generally associated with a lower altitude. As can be seen, the altitude differences for each data cluster **402**, **404** and **406** generally fall relatively close together within their cluster throughout the time of day.

[0046] Additionally, data points that fall outside of the clusters may be considered to be extreme anomalies that should be discarded, instead of included for use in the determination of the HVAC effect. For example, some of the data points for the vehicle that is travelling relatively fast (i.e., the

triangles) might fall above the high cluster **404** or below the low cluster **406**, which might be expected due to the high wind effect of the fast moving vehicle, especially if the vehicle's windows were open, but also to some extent when the windows are closed. In this case, the wind effect could render any quantification of the HVAC effect unreliable, so such extreme data points should preferably be discarded or removed from further calculations. For example, these extreme data points should not be used for calibration purposes. However, there might be situations in which an altitude may have to be determined using such an extreme data point, since there may be no other option. Nevertheless, datapoints taken in a location context of a moving vehicle are preferably discarded before a calibration calculation is performed, because such datapoints are typically noisy. [0047] The data points for the location context of the computing device **102** not being in either a building or a vehicle (i.e., the squares) are all as expected within the middle data cluster **402** for when the computing device **102** is not in an environment that can be subject to an HVAC Effect. On the other hand, the other data points for the building or vehicle (i.e., circles, triangles, and Xs) appear mostly to fall within each data cluster **402**, **404** and **406**. For example, when the HVAC system of the building or vehicle is not blowing air, then there might not be any HVAC pull or push effect, in which case, the data points for the building or vehicle may fall within the middle data cluster **402**. Additionally, when a window of a building or slowly moving vehicle is open, then this can negate any HVAC effect even when the HVAC system is blowing air, in which case, such environments that are open to ambient pressure may be considered “not well-sealed” or “leaky”, and the data points for the building or vehicle may fall within the middle data cluster **402**. On the other hand, when the HVAC system of the building or vehicle is blowing air into an area of the building or vehicle, then there is an HVAC push effect, in which case, the data points for the building or vehicle may fall within the low data cluster **406**. However, when the HVAC system of the building or vehicle is blowing air out of an area of the building or vehicle, then there is an HVAC pull effect, in which case, the data points for the building or vehicle may fall within the high data cluster **404**. Additionally, some buildings or vehicles (or areas or rooms thereof) may experience only an HVAC push effect and not an HVAC pull effect, or vice versa, so only one of the two high/low data clusters **404** and **406** might be seen.

[0048] The data points within the middle data cluster **402** generally represent situations in which the computing device **102** is not in an environment that can be subject to an HVAC Effect (e.g., when outdoors) or is not in an environment that is currently subject to an HVAC Effect (e.g., when the HVAC system is not currently blowing/moving air). The data in the middle data cluster **402** is indicative of overall bias in the altitude difference and can thus reliably be used to calibrate the pressure sensor of the computing device **102**, i.e., calculate a calibration value and a calibration uncertainty value as described, for example, in U.S. Pat. Nos. 11,555,699 and 11,073,441, but no HVAC effect can be determined from this data. (U.S. Pat. Nos. 11,555,699 and 11,073,441 are incorporated herein by reference as if fully set forth herein.) Additionally, the data in the middle data cluster **402** can also reliably be used to determine/estimate an altitude of the computing device **102**. However, to be certain that there were no data points that were subject to HVAC effects, the data points for the location contexts in a building or vehicle (i.e., circles, triangles, and Xs) within the middle data cluster **402** may be excluded from the calibration or altitude calculations. Various conventional calibration techniques can thus be used to obtain a calibration value which, in the illustrated example, would be about 5-6 meters. These calibration and altitude calculations may be done similarly to step **218** in the process **200**, since they are done without HVAC effects.

[0049] At **308**, the presence or magnitude of the HVAC effect is determined based on the middle data cluster **402** (i.e., a first cluster) and the high or low data cluster **404** or **406** (i.e., a second cluster). In other words, the data points in the high data cluster **404** relative to the data points in the middle data cluster **402** can be used to determine or quantify a magnitude of an HVAC pull effect for the building or vehicle (or area or room therein) within which the measurements for the data points were taken. Likewise, the data points in the low data cluster **406** relative to the data points in

the middle data cluster **402** can be used to determine or quantify a magnitude of an HVAC push effect for the building or vehicle (or area or room therein) within which the measurements for the data points were taken. For example, in some embodiments, a first, second and third average, central tendency, or median of the data points in the middle data cluster **402**, the data points in the high data cluster **404**, and the data points in the low data cluster **406**, respectively, are calculated (depending on availability of data for a given building or vehicle). Then the magnitude of the HVAC pull effect for a building or vehicle is determined based on a difference between the first average (for the middle data cluster **402**) and the second average (for the high data cluster **404**), e.g., HVAC pull effect magnitude=first average–second average. Similarly, the magnitude of the HVAC push effect for a building or vehicle is determined based on a difference between the first average (for the middle data cluster **402**) and the third average (for the low data cluster **406**), e.g., HVAC push effect magnitude=first average–third average. The HVAC effect magnitude is then returned to step **214** of the process **200**. If the HVAC effect magnitude (push or pull) is about equal to or greater than a typical height of a building floor (e.g., about ± 3 m) or half this height value, then the HVAC effect could cause the resulting altitude calculation to indicate that the computing device **102** is on the next floor above or below the actual floor. Additionally, a “difference threshold” may be established as an indicator for whether potential HVAC effects are likely to be present and should be taken into account. For example, if the data cluster **404** or **406** is less than the difference threshold (i.e., a relatively small difference value, e.g., about 1-2 m) from the middle data cluster **402**, then the altitude difference may be considered to be due to noise rather than an HVAC effect. Therefore, in some embodiments, the altitude difference for the data cluster **404** or **406** may be used to determine the HVAC effect only when the altitude difference is greater than or equal to the difference threshold.

[0050] In some embodiments, the time of day data is used at **308** to determine different HVAC push/pull effect magnitudes for a building or vehicle that may be likely to occur at different times of day. Similar to the time of day, in some embodiments, time of year data is used at **308** to determine different HVAC push/pull effect magnitudes for a building or vehicle that may be likely to occur at different times of the year.

[0051] In some embodiments, instead of quantifying the HVAC effect, the process **300** merely determines at **308** whether an HVAC effect is likely to be present. Thus, the magnitude need not be calculated in this case. For example, the data points in the clusters **404** and **406** are indicative of the presence of an HVAC effect, regardless of the magnitude thereof. In some embodiments, data points simply being outside of (i.e., above or below) the central/middle data cluster **402** are indicative of the presence of an HVAC effect for these data points. In such cases, at **308**, the process **300** returns a flag or other indication that there is likely an HVAC effect (without a magnitude thereof) to the process **200** at **214**, so that the process **200** can use the mere presence of the HVAC effect to determine (at **216**) not to calibrate the pressure sensor or estimate the altitude of the computing device **102** using the HVAC-affected data, e.g., due to the possibility that the HVAC effect would render the resulting calibration value or altitude too uncertain. In such cases, it may be preferable to use a previously determined calibration value and uncertainty.

[0052] In some embodiments of the pressure-based version of the process **300**, at **302**, the process **300** associates (i.e., maps) the pressure differences (e.g., as determined at **212** for multiple time points) with the corresponding time points and location contexts. The example of FIG. **4** can be used to illustrate how the pressure differences, time points, and location contexts can be associated. As above, the location context is indicated by the circles, triangles, Xs, and squares.

[0053] The pressure difference data is distributed in clusters that are generally the same as the data clusters **402**, **404** and **406** but with reversed descriptions and tighter restrictions on duration length and spatial separations. For example, at **304**, the process **300** determines a first cluster (e.g., a middle data cluster **402**) of the mapped pressure differences for when the computing device **102** is not in an environment that can be subject to an HVAC Effect. Additionally, at **306**, the process **300**

determines one or more second clusters (e.g., data clusters **404** and **406**) of the mapped pressure differences for when the computing device **102** is in an environment that can be subject to an HVAC Effect. The data cluster **404** is a high cluster that represents a potential HVAC push effect, since an HVAC push results in a higher air pressure, which is generally associated with a lower altitude. Thus, when the HVAC system of the building or vehicle is blowing air into an area of the building or vehicle, then there is an HVAC push effect, in which case, the data points for the building or vehicle may fall within the high data cluster **404**. Additionally, the data cluster **406** is a low cluster that represents a potential HVAC pull effect, since an HVAC pull results in a lower air pressure, which is generally associated with a higher altitude. Thus, when the HVAC system of the building or vehicle is blowing air out of an area of the building or vehicle, then there is an HVAC pull effect, in which case, the data points for the building or vehicle may fall within the low data cluster **406**. The pressure differences for each data cluster **402**, **404** and **406** generally fall relatively close together within their cluster throughout the time of day.

[0054] As above, data points that fall outside of the clusters may be considered to be extreme anomalies that should preferably be discarded, instead of included for use in the determination of the HVAC effect or a calibration.

[0055] Similar to the description above, at **308**, the presence or magnitude of the HVAC effect is determined based on the middle data cluster **402** (i.e., a first cluster) and the high or low data cluster **404** or **406** (i.e., a second cluster). In other words, the data points in the high data cluster **404** relative to the data points in the middle data cluster **402** can be used to determine or quantify a magnitude of an HVAC push effect for the building or vehicle (or area or room therein) within which the measurements for the data points were taken. Likewise, the data points in the low data cluster **406** relative to the data points in the middle data cluster **402** can be used to determine or quantify a magnitude of an HVAC pull effect for the building or vehicle (or area or room therein) within which the measurements for the data points were taken.

[0056] FIG. 5 is a simplified flowchart of an example process **500** for determining (e.g., at step **204** in FIG. 2) whether the computing device **102** is in a building, in accordance with some embodiments. The particular steps, combination of steps, and order of the steps for this process are provided for illustrative purposes only. Other processes with different steps, combinations of steps, or orders of steps can also be used to achieve the same or similar result. Features or functions described for one of the steps performed by one of the components may be enabled in a different step or component in some embodiments. Additionally, some steps may be performed before, after or overlapping other steps, in spite of the illustrated order of the steps.

[0057] At **502**, the process **500** obtains the two dimensional (i.e., 2D or horizontal) location of the computing device **102**. For example, the location determined at **202** may be used.

[0058] At **504**, one or more of several techniques, tests or options may be used to determine if it is likely that the computing device **102** is inside a building (e.g., building **108** or **110** in FIG. 1). If any of these techniques indicates (as determined at **506**) that the location context of the computing device **102** is likely to be in a building, then the process **500** proceeds to **508** to return an indicator to the process **200** (at **204**) that the location context is in a building. On the other hand, if none of these techniques indicates (at **506**) that the location context of the computing device **102** is likely to be in a building, then the process **500** proceeds to **510** to return an indicator to the process **200** (at **204**) that the location context is not in a building.

[0059] In some embodiments, for example, the 2D location (e.g., the latitude and longitude and optionally 2D uncertainty) of the computing device **102** are mapped to a terrain or building database that provides the locations of buildings in the general area of the 2D location. This mapping would, thus, indicate whether the computing device **102** is in the building. The 2D location and uncertainty, for example, might indicate that the computing device **102** is within a location area **134** shown in FIG. 1. A large portion of the location area **134** maps to the building **110** (e.g., a key ratio between the device footprint of the computing device **102** and the building

footprint of the building **110** is above an appropriate threshold value), so there is a high likelihood that the computing device **102** is within this building, e.g., at **126**. In this case, the process **500** branches to **508** to return to the process **200** (at **204**) that the location context is in a building. [0060] On the other hand, the 2D location and uncertainty, for example, might indicate that the computing device **102** is within a location area **136** shown in FIG. 1 (e.g., when the computing device **102** is at **130**). No portion of the location area **136** maps to a building, so there is a low likelihood that the computing device **102** is within any building. In this case, the process **500** branches to **510** to return to the process **200** (at **204**) that the location context is not in a building. [0061] In contrast, the 2D location and uncertainty, for example, might indicate that the computing device **102** is within a location area **138** shown in FIG. 1. A first portion of the location area **138** maps to the building **110** (as if the computing device **102** was at **126**), and second portion of the location area **138** maps to the building **108** (as if the computing device **102** was at **124**), but a third portion of the location area **138** does not map to any building (as if the computing device **102** was outdoors at **132**). Depending on the key ratio between the device footprint and the building footprint of either of the buildings **108** or **110**, the process **500** makes the appropriate determination of whether the computing device is within one of the buildings **108** or **110** or is not within any building and returns (at **508** or **510**) to the process **200** (at **204**) the appropriate location context that the computing device **102** is in a building or not in a building. Additionally, in some embodiments, the location context may specify which building the computing device **102** is in, because different buildings could have different HVAC effects. As illustrated, the portion of the location area **138** that does not map to any building appears to be much larger than either of the other portions, so in this example the process **500** returns (at **510**) to the process **200** (at **204**) the location context that the computing device **102** is not in a building.

[0062] In a situation where a user (with the computing device **102**) is standing on a rooftop of a building or parking garage, the environment would be a non-HVAC environment, since it would be open to the ambient environment. In some embodiments, therefore, a 3D location including an initial altitude determination (and optionally 3D uncertainty) are made at **502**, so that the 2D location and the possible altitude are taken into account (e.g., within given horizontal and vertical thresholds) in determining whether the computing device **102** is inside a building.

[0063] In some embodiments, the process **500** determines (at **504**) a proximity or connection of the computing device to a wireless access point (WAP) **140** (e.g., a Wi-Fi transceiver or router), the location of which is known to be in a building, because a wireless connection to the WAP would indicate that the computing device **102** is within the building. In this case, if the WAP **140** is known to be inside a multi-floor building (as shown in building **110**), and the process **500** determines that the computing device **102** is wirelessly connected to the WAP **140**, then the process **500** further determines that the computing device **102** is inside the building **110**. Thus, the process **500** branches to **508** to return to the process **200** (at **204**) that the location context is in a building.

[0064] In some embodiments, the process **500** determines (at **504**) a GNSS signal strength, e.g., of the positioning signal **116** from the satellite **118**. In this case, when the strength of the satellite signal is attenuated from the level that it should be if the computing device **102** were outdoors with a direct line-of-sight access to the satellite **118**, then this indicates that the computing device **102** is not outdoors with a direct line-of-sight access to the satellite **118**, so the process **500** determines that the computing device **102** is likely to be indoors and, thus, likely to be in a building, e.g., at **124** in building **108** or at **126** in building **110**. Thus, the process **500** branches to **508** to return to the process **200** (at **204**) that the location context is in a building.

[0065] In some embodiments, the process **500** performs each of the above-described techniques (among others) for determining if the computing device **102** is in a building. If any one of these techniques indicates that the computing device **102** is in a building, then the process **500** branches to **508** to return to the process **200** (at **204**) that the location context is in a building. However, if none of these techniques indicates that the computing device **102** is in a building, then the process

500 branches to **510** to return to the process **200** (at **204**) that the location context is not in a building.

[0066] FIG. **6** is a simplified flowchart of an example process **600** for determining (e.g., at step **204** in FIG. **2**) whether the computing device **102** is in a vehicle, in accordance with some embodiments. The particular steps, combination of steps, and order of the steps for this process are provided for illustrative purposes only. Other processes with different steps, combinations of steps, or orders of steps can also be used to achieve the same or similar result. Features or functions described for one of the steps performed by one of the components may be enabled in a different step or component in some embodiments. Additionally, some steps may be performed before, after or overlapping other steps, in spite of the illustrated order of the steps.

[0067] At **602**, the process **600** obtains the two dimensional (i.e., 2D or horizontal) location of the computing device **102**. For example, the location determined at **202** may be used.

[0068] At **604**, one or more of several techniques, tests or options may be used to determine if it is likely that the computing device **102** is inside a vehicle (e.g., at **122** or **128** in FIG. **1**). For example, if measurements from the pressure sensor of the computing device **102** over a relatively short period of time are relatively noisy, i.e., exhibit relatively large differences, then it can be inferred that the computing device **102** is in a rapidly moving vehicle. On the other hand, if the differences between such pressure measurements are below an appropriate noise threshold, then it may be difficult to distinguish between walking, biking, and some traffic activities. Therefore, the techniques used at **604** to determine if it is likely that the computing device **102** is in a vehicle may each provide different levels of confidence in the resulting determination. For example, if the location obtained at **602** maps to a location that is likely not to have a vehicle (e.g., hilly hiking or biking trails), then the confidence level in the determination that the computing device **102** is not in a vehicle may be higher. If any of these techniques indicates (as determined at **606**) that the location context of the computing device **102** is likely to be in a vehicle, then the process **600** proceeds to **608** to return an indicator to the process **200** (at **204**) that the location context is in a vehicle. On the other hand, if none of these techniques indicates (as determined at **606**) that the location context of the computing device **102** is likely to be in a vehicle, then the process **600** proceeds to **610** to return an indicator to the process **200** (at **204**) that the location context is not in a vehicle.

[0069] In some embodiments, for example, the 2D location (e.g., the latitude and longitude and optionally 2D uncertainty) of the computing device **102** are mapped to a terrain, road, bridge, parking lot, or other similar database that provides the locations of roads, bridges, tunnels, parking lots, etc. in the general area of the 2D location. Thus, the 2D location and uncertainty, for example, might indicate that the computing device **102** is within a location area **142** shown in FIG. **1**. A large portion of the location area **142** maps to a road, so there is a high likelihood that the computing device **102** is in a vehicle on this road, e.g., at **122**. Additionally, if the road is a divided-lane highway, then the vehicle is likely to be moving relatively fast; but if the road is a residential side street, then the vehicle is likely to be moving relatively slowly. (A previous or subsequent location/time datapoint (before or after the datapoint being analyzed by the process **600**) can be used to confirm the speed of the vehicle if the time difference between the two datapoints is relatively small. Additionally, other information can be used to infer the speed of the vehicle, e.g., a database of speed limits, proximity to building types such as schools or hospitals, among others.) In this case, the process **600** branches to **608** to return to the process **200** (at **204**) that the location context is in a vehicle.

[0070] In some embodiments, in another example, the 2D location and uncertainty, for example, might indicate that the computing device **102** is within a location area **144** shown in FIG. **1**, as determined based on the terrain, road, bridge, parking lot, or other similar database. A large portion of the location area **144** maps to a parking lot, so there is a high likelihood that the computing device **102** is in a vehicle on this parking lot, e.g., at **128**. If the computing device **102** is not

moving (still), then the process **600** may determine that the computing device **102** is in the vehicle while it is parked. In this case, the process **600** branches to **608** to return to the process **200** (at **204**) that the location context is in a vehicle. If the computing device **102** is moving slowly (but faster than a typical walking speed), then the process **600** may determine that the computing device **102** is in the vehicle while it is driving through the parking lot. In this case, the process **600** branches to **608** to return to the process **200** (at **204**) that the location context is in a vehicle. If the computing device **102** is moving at a typical walking speed, does not follow vehicle lanes of the parking lot, and/or moves in a straight line (across a winding parking lot), then the process **600** may determine that the computing device **102** is no longer in the vehicle but is now being carried by the user through the parking lot. In this case, the process **600** branches to **610** to return to the process **200** (at **204**) that the location context is not in a vehicle.

[0071] In some embodiments, in another example, the 2D location and uncertainty, for example, might indicate that the computing device **102** is within the location area **136** shown in FIG. 1 (e.g., when the computing device **102** is at **130**). No portion of the location area **136** maps to a road, bridge, tunnel, parking lot, etc., so there is a low likelihood that the computing device **102** is within any vehicle. In this case, the process **600** branches to **610** to return to the process **200** (at **204**) that the location context is not in a vehicle.

[0072] In some embodiments, the process **600** determines or identifies that the computing device **102** is communicatively connected to a vehicle via wired or wireless means as may be indicated by an application running on the computing device **102**, which indicates that the computing device **102** is within the vehicle. Thus, the process **600** determines that there is a high likelihood that the computing device **102** is inside the vehicle to which it is communicatively connected. Therefore, the process **600** branches to **608** to return to the process **200** (at **204**) that the location context is in a vehicle.

[0073] In some embodiments, the process **600** determines that the computing device **102** has traversed a relatively large distance in a relatively short amount of time, i.e., the computing device **102** is moving relatively fast, which indicates that the computing device **102** is likely to be in a vehicle that is moving relatively fast, i.e., faster than a typical walking, skating, or biking speed. When this situation occurs, therefore, the process **600** branches to **608** to return to the process **200** (at **204**) that the location context is in a vehicle. This option may be done if at least two datapoints are relatively close together in time.

[0074] In some embodiments, the process **600** determines that the computing device **102** is changing cell tower connections. Although this can happen when the user is travelling on foot, by bicycle, or other non-vehicle means, the speeds and distances travelled by vehicle make it more likely that a cell tower connection change is indicative of the computing device **102** being in a moving vehicle. When this situation occurs, therefore, the process **600** branches to **608** to return to the process **200** (at **204**) that the location context is in a vehicle.

[0075] In some embodiments, the process **600** obtains the activity context from a sensor hub of the computing device **102**. This can be done with computing devices that determine their activity context with an app that is independent from the features described herein. Typical activity contexts that such computing devices track include walking, biking, driving, and standing still, among others. Therefore, when the activity context obtained from the sensor hub is “driving” along with an activity context confidence level that is above an activity confidence threshold (e.g., about 60% or greater), then the process **600** branches to **608** to return to the process **200** (at **204**) that the location context is in a vehicle. On the other hand, when the activity context obtained from the sensor hub is “walking”, “biking” or other specifically non-vehicle activity, then the process **600** branches to **610** to return to the process **200** (at **204**) that the location context is not in a vehicle. Other activity contexts, such as standing still, could be ambiguous with respect to whether the computing device is in a vehicle.

[0076] In some embodiments, the process **600** determines if the location or location area of the

computing device **102** maps to a nearby building (e.g., within a given threshold), but the key ratio between the device footprint and the building footprint is too small for the process **500** (FIG. 5) to determine that the computing device **102** is inside the building. In this situation, the computing device **102** might be in a parking lot outside the building, which further indicates that the computing device **102** is likely to be in a vehicle, so the process **600** branches to **608** to return to the process **200** (at **204**) that the location context is in a vehicle.

[0077] In some embodiments, the process **600** performs each of the above-described techniques (among others) for determining if the computing device **102** is in a vehicle. If any one of these techniques indicates that the computing device **102** is in a vehicle, then the process **600** branches to **608** to return to the process **200** (at **204**) that the location context is in a vehicle. However, if none of these techniques indicates that the computing device **102** is in a vehicle, then the process **600** branches to **610** to return to the process **200** (at **204**) that the location context is not in a vehicle.

[0078] In some situations, when the computing device **102** is in a vehicle moving at a relatively high rate of speed, the environment in the vehicle is likely to be unstable with respect to air pressure due to wind turbulence. In this case, any potential HVAC effect would be difficult to disentangle or distinguish from the turbulent pressures caused by the high wind effects. Therefore, datapoints from such situations should preferably be denoted as “in vehicle, likely at high rate of speed” and flagged to not be used in any analysis for HVAC effect. U.S. Pat. No. 11,073,441, which is incorporated herein by reference as if fully set forth herein, describes such situations and metrics for the pressure variation effects thereof.

[0079] FIGS. 7-9 describe a process **700** for determining (e.g., at step **204** in FIG. 2) if it is likely that the computing device **102** is in almost any type of environment that can be subject to an HVAC effect, in accordance with some embodiments. The process **700** does this using temperature data and without determining the location context (e.g., in a building or vehicle) of the computing device **102**. Instead, location context is inferred from temperature measurements and variations to be any unspecified HVAC environment (or not). Thus, if the location context of the computing device **102** (as determined by process **500** or **600**) is not considered reliable or is not available, it may be possible to use the process **700** instead of or in addition to either process **500** or **600** to determine whether the computing device **102** is in an HVAC environment. The particular steps, combination of steps, and order of the steps for this process are provided for illustrative purposes only. Other processes with different steps, combinations of steps, or orders of steps can also be used to achieve the same or similar result. Features or functions described for one of the steps performed by one of the components may be enabled in a different step or component in some embodiments. Additionally, some steps may be performed before, after or overlapping other steps, in spite of the illustrated order of the steps.

[0080] At **702**, the process **700** determines the temperature of the computing device **102** (device temperature) based on a measurement by a temperature sensor therein. In some embodiments, to account for heat generated by the computing device **102**, the measured temperature is adjusted by an offset in order to align closer with true ambient temperature of the current environment of the computing device **102**. This adjustment may be based upon device specifications, lab experimentation, or user input (e.g., subtract 10 degrees C. or 10% of the battery sensor temperature to arrive at the estimated device temperature).

[0081] At **704**, the process **700** obtains a reference temperature (e.g., an outdoor temperature sourced from a network of reference sensors, a National Weather Service data feed, etc.) based on the location of the computing device **102**. In some embodiments, the reference temperature may optionally be adjusted by a conventional formula using a thermal lapse rate to account for the difference between an initially determined altitude of the computing device **102** and the altitude of the reference sensor. In some embodiments, the reference temperature may be considered to be the same at both altitudes.

[0082] At **706**, the process **700** compares the device temperature and the reference temperature

based on example guidelines for techniques, tests or options illustrated by a graph **800** in FIG. **8**. FIG. **8** is a simplified graph **800** of the reference temperature vs. the device temperature illustrating examples of possible or likely location contexts and changes in location contexts for use in the process **700**, in accordance with some embodiments. The graph **800** includes several regions **802-828** that provide an indication of whether the computing device **102** is likely to be indoors (i.e., in an HVAC environment), outdoors (i.e., in a non-HVAC environment), or in an uncertain environment. The particular regions **802-828** and the temperature borders thereof are shown for illustrative and explanatory purposes only, because other embodiments could provide borders that are somewhat different, not include some of the regions, or include different regions that serve generally the same purpose.

[0083] Additionally, FIG. **9** shows simplified graphs **902-908** of the reference temperature (**902** and **906**) and the device temperature (**904** and **908**) vs. time of day illustrating relative differences in the reference temperature and the device temperature depending on location contexts and changes in location contexts for interpreting the data shown in FIG. **8**, in accordance with some embodiments. In particular, FIG. **9** shows simulated time profiles of two devices transitioning over a period of a few minutes from outdoor environments to indoor comfortable environments, where an HVAC system could potentially affect the indoor environments. In these examples, it takes about 20 minutes for the temperature of the computing device **102** to equalize with the new indoor environment.

[0084] When the process **700** determines (at **708**) that the temperature comparison at **706** (in accordance with the example guidelines of FIG. **8**) indicates that the computing device **102** is likely to be in a location context that is in an environment that can be subject to an HVAC effect, then the process **700** branches to **710** to return to the process **200** (at **204**) that the location context is an HVAC environment. On the other hand, when the process **700** determines (at **708**) that the temperature comparison at **706** (in accordance with the example guidelines of FIG. **8**) indicates that the computing device **102** is likely to be in a location context that is not in an environment that can be subject to an HVAC effect, then the process **700** branches to **712** to return to the process **200** (at **204**) that the location context is a non-HVAC environment. Alternatively, when the process **700** determines (at **708**) that the temperature comparison at **706** (in accordance with the example guidelines of FIG. **8**) indicates that it is uncertain whether the computing device **102** is likely to be in a location context that is or is not in an environment that can be subject to an HVAC effect, then the process **700** branches to **714** to return to the process **200** (at **204**) that the location context is in an uncertain environment (i.e., uncertain to be in an environment that can be subject to an HVAC effect).

[0085] Regions **802**, **803**, **804**, **805** and **806** of the graph **800** are on a diagonal in which the reference temperature and the device temperature are about the same or within a threshold temperature range from each other, e.g., within about ± 5 degrees C. of each other. The region **802**, for example, represents relatively cold temperatures for the reference temperature and the device temperature, e.g., less than an appropriate indoor “cold” threshold temperature or about 15 degrees C. (± 5 degrees C.) and greater than a high end of a typical freezer (e.g., a walk-in freezer) temperature range (e.g., about -15 to -20 degrees C.). (The indoor cold threshold temperature is generally at a low end of a temperature range that people typically set for a comfortable indoor environment at home or work.) The reference temperature graph **906** and the device temperature graph **908** illustrate an example of this situation in portion **910** wherein these two graphs **906** and **908** are very close to each other at a relatively low temperature of about 10 degrees C. Additionally, the region **803** also represents relatively cold temperatures for the reference temperature and the device temperature that are not only less than the appropriate indoor “cold” threshold temperature but also less than a low end of the typical freezer temperature range. A situation in which both the reference temperature and the device temperature are relatively cold (but not potentially explained by being in a freezer as described below) is most likely to occur

when the computing device **102** is outside any enclosed environment, i.e., in a non-HVAC environment, during cold weather long enough for the computing device **102** also to become cold. When this situation occurs, therefore, the process **700** determines (at **708**) that the temperature comparison at **706** indicates that the computing device **102** is likely to be in a location context that is not in an environment that can be subject to an HVAC effect, so the process **700** branches to **712** to return to the process **200** (at **204**) that the location context is a non-HVAC environment.

[0086] The region **805**, on the other hand, represents an example situation that can be a potential exception to the situations represented by regions **802** and **803**. For the situation of region **805**, the reference temperature and the device temperature are about the same or within the threshold temperature range from each other and are less than the appropriate indoor “cold” threshold temperature (like regions **802** and **803**) but are also within the typical freezer temperature range (± 5 degrees C.) (unlike regions **802** and **803**). The situation of region **805** could occur under the same circumstances as those of regions **802** and **803** or when the computing device **102** is within a freezer (e.g., when the user is working inside a walk-in freezer) and the reference temperature is also at about the same temperature range. Thus, the situation of region **805** could occur when the computing device **102** is either outside or inside. In this case, the process **700** cannot confidently determine whether the location context of the computing device **102** is in an HVAC environment. When this situation occurs, therefore, the process **700** returns (at **714**) to the process **200** (at **204**) that the location context is in an uncertain environment (i.e., uncertain to be in an environment that can be subject to an HVAC effect) or alternatively branches to **712** to return to the process **200** (at **204**) that the location context is a non-HVAC environment (as a default result).

[0087] In another example, the region **806** illustrates another situation in which the reference temperature and the device temperature are about the same or within the threshold temperature range from each other, e.g., within about ± 5 degrees C. of each other, but within a relatively hot temperature range. The relatively hot temperature range may be, for example, greater than an appropriate indoor “hot” threshold temperature or about 35 degrees C. (± 5 degrees C.). (The indoor hot threshold temperature is generally at a high end of a temperature range that people typically set for a comfortable indoor environment at home or work.) The reference temperature graph **902** and the device temperature graph **904** illustrate an example of this situation in portion **912** wherein these two graphs **902** and **904** are very close to each other at a relatively high temperature. A situation in which both the reference temperature and the device temperature are relatively hot is most likely to occur when the computing device **102** is outside any enclosed environment, i.e., a non-HVAC environment, during hot/warm weather long enough for the computing device **102** also to become hot/warm. When this situation occurs, therefore, the process **700** determines (at **708**) that the temperature comparison at **706** indicates that the computing device **102** is likely to be in a location context that is not in an environment that can be subject to an HVAC effect, so the process **700** branches to **712** to return to the process **200** (at **204**) that the location context is a non-HVAC environment.

[0088] In another example, the region **804** illustrates another situation in which the reference temperature and the device temperature are about the same or within the threshold temperature range from each other, e.g., within about ± 5 degrees C. of each other, but within a relatively comfortable temperature range for a typical user of the computing device **102**. The illustrated temperature range of about 15 degrees C. to 35 degrees C. (i.e., 25 degrees C. ± 10 degrees C.) is not only a common outdoor temperature range but also a common temperature range at which indoor environments (e.g., inside homes, office buildings, etc.) are typically maintained by an HVAC system for the comfort of the people therein, i.e., the relatively comfortable temperature range is greater than the appropriate indoor “cold” threshold temperature and less than the appropriate indoor “hot” threshold temperature. Thus, a situation in which both the reference temperature and the device temperature are within a relatively comfortable temperature range could occur when the computing device **102** is either outside or inside. In this case, the location context

of the computing device **102** is in an uncertain environment. When this situation occurs, therefore, the process **700** returns (at **714**) to the process **200** (at **204**) that the location context is in an uncertain environment (i.e., uncertain to be in an environment that can be subject to an HVAC effect) or alternatively branches to **712** to return to the process **200** (at **204**) that the location context is a non-HVAC environment (as a default result).

[0089] In another example, the region **808** illustrates a situation in which the device temperature is within the relatively comfortable temperature range (e.g., about 15 degrees C. to 35 degrees C.), but the reference temperature is warmer than the device temperature, e.g., by more than the threshold temperature range. The reference temperature graph **902** and the device temperature graph **904** illustrate an example of this situation in portion **914** wherein the reference temperature graph **902** is at a relatively high temperature and the device temperature graph **904** is at a relatively mid-level temperature. In such a situation, the temperature of the computing device **102** is apparently not affected by the outdoor temperature. This situation is likely to occur when the computing device **102** is inside a temperature-controlled environment, e.g., with the air-conditioning (A/C) or cooling turned on. When this situation occurs, therefore, the process **700** branches to **710** to return to the process **200** (at **204**) that the location context is an HVAC environment.

[0090] In another example, the region **810** illustrates a situation in which the device temperature is within the relatively comfortable temperature range (e.g., about 15 degrees C. to 35 degrees C.), but the reference temperature is colder than the device temperature. The reference temperature graph **906** and the device temperature graph **908** illustrate an example of this situation in portion **916** wherein the reference temperature graph **906** is at a relatively low temperature and the device temperature graph **908** is at a relatively mid-level temperature. In such a situation, the temperature of the computing device **102** is apparently not affected by the outdoor temperature. This situation is likely to occur when the computing device **102** is inside a temperature-controlled environment, e.g., with the heater turned on to maintain the relatively comfortable temperature range. When this situation occurs, therefore, the process **700** branches to **710** to return to the process **200** (at **204**) that the location context is an HVAC environment.

[0091] In another example, the region **812** (shown to apparently overlap regions **802** and **810**) illustrates a situation in which the reference temperature is relatively cold and the device temperature is not quite the same as the reference temperature (as for region **910**) but also not quite within the relatively comfortable temperature range (as for region **916**). For example, the reference temperature is below or colder than the device temperature. Also, the device temperature is 1) above the reference temperature and 2) at or near a preferred low-end temperature (e.g., the appropriate “cold” threshold temperature or about 15 C) of the comfortable temperature range within plus-or-minus an acceptable low-end temperature deviation amount (e.g., +/-5 degrees C. or 10-20 degrees C.). The reference temperature graph **906** and the device temperature graph **908** illustrate an example of this situation in portion **918** (e.g., a low-to-high transition temperature range) wherein the reference temperature graph **906** is at a relatively low temperature and the device temperature graph **908** appears to be transitioning from the relatively low temperature to the mid-level temperature. This situation is likely to occur when the computing device **102** was recently exposed to a relatively cold outdoor environment but has moved into a warmer indoor environment and is now slowly warming up to match the indoor environment. The indoor environment is, thus, likely to be an HVAC environment that is temperature-controlled with the heater turned on to maintain the relatively comfortable temperature range. When this situation occurs, therefore, the process **700** branches to **710** to return to the process **200** (at **204**) that the location context is an HVAC environment. Alternatively, the process **700** branches to **714** to return to the process **200** (at **204**) that the location context is in an uncertain environment (i.e., uncertain to be in an environment that can be subject to an HVAC effect), because the computing device **102** could be in an environment like that of region **816** (described below) which could be outside, in an

environment like that of region **810** which could be inside, or could be transitioning between environments of regions **810** and **816**. Selecting between branching to **710** or to **714** could depend on the number of datapoints or temperature measurements and their spread within region **812** and immediately outside thereof. For example, more datapoints spread out within region **812** and immediately outside the region **812** (e.g., within region **810**) could result in greater certainty that the computing device **102** has transitioned from outside (region **816**) to inside (region **810**) and is then stabilizing to thermal equilibrium with the enclosed temperature-controlled environment of region **810**.

[0092] In another example, the region **814** (shown to apparently overlap regions **806** and **808**) illustrates a situation in which the reference temperature is relatively hot/warm and the device temperature is not quite the same as the reference temperature (as for region **912**) but also not quite within the relatively comfortable temperature range (as for region **914**). For example, the reference temperature is above or warmer than the device temperature. Also, the device temperature is 1) below the reference temperature and 2) at or near a preferred high-end temperature (e.g., the appropriate “hot” threshold temperature or about 35 C) of the comfortable temperature range plus-or-minus an acceptable high-end temperature deviation amount (e.g., ± 5 degrees C. or 30-40 degrees C.). The reference temperature graph **902** and the device temperature graph **904** illustrate an example of this situation in portion **920** (e.g., a high-to-low transition temperature range) wherein the reference temperature graph **902** is at a relatively high temperature and the device temperature graph **904** appears to be transitioning from the relatively high temperature to the mid-level temperature. This situation is likely to occur when the computing device **102** was recently exposed to a relatively hot/warm outdoor environment but has moved into a cooler indoor environment and is now slowly cooling down to match the indoor environment. The indoor environment is, thus, likely to be an HVAC environment that is temperature-controlled with the A/C cooling turned on to maintain the relatively comfortable temperature range. When this situation occurs, therefore, the process **700** branches to **710** to return to the process **200** (at **204**) that the location context is an HVAC environment. Alternatively, the process **700** branches to **714** to return to the process **200** (at **204**) that the location context is in an uncertain environment (i.e., uncertain to be in an environment that can be subject to an HVAC effect), because the computing device **102** could be in an environment like that of region **818** (described below) which could be outside, in an environment like that of region **808** which could be inside, or could be transitioning between environments of regions **808** and **818**. Selecting between branching to **710** or to **714** could depend on the number of datapoints or temperature measurements and their spread within region **814** and immediately outside thereof. For example, more datapoints spread out within region **814** and immediately outside the region **814** (e.g., within region **808**) could result in greater certainty that the computing device **102** has transitioned from outside (region **818**) to inside (region **808**) and is then stabilizing to thermal equilibrium with the enclosed temperature-controlled environment of region **808**.

[0093] In another example, the region **816** illustrates a situation in which the reference temperature is relatively cold and the device temperature is not quite the same as the reference temperature (as for region **910**) but also not clearly within the low-to-high transition temperature range (as for region **918**). For example, the reference temperature is below or colder than the device temperature. Also, the device temperature is 1) above the reference temperature, 2) significantly below the preferred low-end temperature (e.g., the appropriate indoor “cold” threshold temperature or about 15 C) of the comfortable temperature range by at least the acceptable low-end temperature deviation amount (e.g., less than about 10 C, which is about 5 degrees C. below about 15 degrees C.), and 3) above the high end of the typical freezer temperature range. This situation is likely to occur 1) when the computing device **102** is outside (e.g., similar to region **802**) but the computing device **102** is slightly warmed (e.g., by body heat from the user) or 2) when the computing device **102** has entered a temperature-controlled environment like that of region **810** and the temperature

of the computing device **102** is transitioning from the outside cold temperature (or a walk-in freezer cold temperature) to the inside warmer temperature. When this situation occurs, therefore, the process **700** branches to **714** to return to the process **200** (at **204**) that the location context is an uncertain environment.

[0094] In another example, the region **818** illustrates a situation in which the reference temperature is relatively hot/warm and the device temperature is not quite the same as the reference temperature (as for region **912**) but also not clearly within the high-to-low transition temperature range (as for region **920**). For example, the reference temperature is above or warmer than the device temperature. Also, the device temperature is 1) below the reference temperature and 2) significantly above the preferred high-end temperature (e.g., the appropriate “hot” threshold temperature or about 35 C) of the comfortable temperature range by at least the acceptable high-end temperature deviation amount (e.g., greater than about 40 C, which is about 5 degrees C. above about 35 degrees C.). This situation is likely to occur when the computing device **102** is 1) outside in shade (and the reference sensor providing the reference temperature is not shaded), 2) inside but being heated by the sun (e.g., next to a window), or 3) inside but recently transitioned from outside (and the device temperature has just started to transition from the hot outside temperature to the cooler inside temperature). When this situation occurs, therefore, the process **700** branches to **714** to return to the process **200** (at **204**) that the location context is an uncertain environment.

[0095] In another example, the region **820** illustrates a situation in which the device temperature is cooler than (potentially a lot cooler than) the reference temperature. This situation is very unlikely to occur because it is considered to be an unphysical condition for the computing device **102** to be cooler than the ambient temperature due to the fact that there should be at least some additional heating of the computing device **102** by its own active circuitry or the user's body heat. Therefore, it is unlikely that the process **700** would ever encounter this situation and the current datapoint could be erroneous. On the other hand, the situation of region **820** could occur when the computing device **102** has been in an indoor freezer or refrigerator environment and has recently transitioned to a warmer normal temperature-controlled indoor environment like that for regions **808** or **810** or a warmer outdoor environment and the device temperature is in the process of transitioning to the warmer temperature. Alternatively, the opposite situation could have occurred in which the computing device **102** has recently transitioned to the indoor freezer or refrigerator environment. When this situation occurs, therefore, the process **700** branches to **714** to return to the process **200** (at **204**) that the location context is an uncertain environment. Alternatively, if this situation were to occur, then the process **700** would return an error (not shown) to the process **200** (at **204**), and the process **200** would discard the current datapoint as likely being erroneous. Determining whether to label the location context as in an uncertain environment or the current datapoint as being erroneous can depend on whether the location context for a datapoint immediately prior to the current datapoint was for a walk-in freezer or refrigerator.

[0096] In another example, the region **822** illustrates a situation in which the device temperature is warmer than (potentially a lot warmer than) the reference temperature. This situation is likely to occur when the computing device **102** is warmed by direct sunlight, which can potentially significantly increase the temperature of the computing device **102**. This can occur not only when the computing device **102** is outdoors but also when it is indoors in a building or in a vehicle and sitting next to a window through which sunlight can hit it. When this situation occurs, therefore, the process **700** would return an error (not shown) to the process **200** (at **204**), and the process **200** would discard the current datapoint as likely being unreliable.

[0097] In another example, the region **824** illustrates a situation in which the device temperature is within the typical freezer (e.g., a walk-in freezer) temperature range and the reference temperature is not approximately the same as the device temperature (i.e., not within the threshold temperature range from each other). This situation is likely to occur when the computing device **102** has been in a freezer environment (i.e., a type of HVAC environment) long enough to reach equilibrium with

the freezer temperature. When this situation occurs, therefore, the process **700** branches to **710** to return to the process **200** (at **204**) that the location context is an HVAC environment. Additionally, a similar region (not shown) could be defined for a refrigerator environment having a typical refrigerator (e.g., a walk-in refrigerator) temperature range, which would be colder than the comfortable temperature range of regions **808** and **810** but above freezing (i.e., above zero degrees C.).

[0098] In another example, the region **826** illustrates a situation in which the device temperature is cooler than (potentially a lot cooler than) the reference temperature, even cooler than the typical freezer temperature range. This situation is very unlikely to occur because it is considered to be an unphysical condition for the computing device **102** to be cooler than the ambient temperature due to the fact that there should be at least some additional heating of the computing device **102** by its own active circuitry or the user's body heat. Therefore, it is unlikely that the process **700** would ever encounter this situation. Nevertheless, if this situation were to occur, then the process **700** would return an error (not shown) to the process **200** (at **204**), and the process **200** would discard the current datapoint as likely being erroneous.

[0099] In another example, the region **828** illustrates a situation in which the reference temperature is relatively cold and the device temperature is 1) slightly above the reference temperature, 2) significantly below the preferred low-end temperature of the comfortable temperature range, and 3) below the low end of the typical freezer temperature range. This situation is likely to occur when the computing device **102** is outside but the computing device **102** is slightly warmed (e.g., by body heat from the user). When this situation occurs, therefore, the process **700** branches to **710** to return to the process **200** (at **204**) that the location context is a non-HVAC environment.

[0100] FIG. **8** also shows an example of five datapoints (**t0-t4** with lower case "t") for the computing device **102** moving from an outdoor environment to an indoor environment. This example illustrates a path of the computing device **102** as it is transitioning from a cold outside temperature to an indoor, comfortable temperature, potentially subject to HVAC effects. This example shows how changes in the device temperature compared with a reference temperature over a given period of time can indicate the location context of the computing device **102**. The direction of the path provides confidence that the computing device **102** enters an HVAC environment at the later datapoints. The reference temperature graph **906** and the device temperature graph **908** of FIG. **9** is an example time profile for the temperature change that could occur when transitioning through the datapoints of times **t0-t4**.

[0101] The datapoint at time **t0** is within the region **802**, so the computing device **102** was likely to have been outside at this time, i.e., in a non-HVAC environment. The datapoint at time **t1** is on the edge between the region **802** and **804**, so the location context is uncertain for this time. The datapoint **t2** is on the edge between the region **804** and the region **810**, so again the location context is uncertain for this time. The datapoint at time **t3** is within the region **810**, so the location context is likely to be in a HVAC environment. The datapoint at time **t4** is even further within the region **810** than the datapoint at **t3** is, so the location context is even more likely to be in a HVAC environment.

[0102] The datapoints (**t0**, **t3** and **t4**) that provide a likely location context, if taken in isolation from each other, would result in a possible uncertainty level or value for the location context determination. However, it is clear that the device temperature is getting warmer (as time passes from **t0** to **t4**), while the reference temperature is staying cold. In this situation, the process **700** determines that the computing device **102** transitioned from a likely non-HVAC environment (at **t0**) through an uncertain environment (**t1** and **t2**) to a likely HVAC environment (at **t3** and **t4**). This transition provides evidence that the location context determinations for the datapoints **t0**, **t3** and **t4** may have a lower uncertainty or greater confidence level, as calculated by the process **700**, because it appears to be clear that the computing device **102** is warming up (from **t0** to **t4**) due to having been outside in the cold and then transitioning to inside where it was warmer. The actual point of

the transition from the non-HVAC environment to the HVAC environment is likely to be at some point around time **t1** or time **t2**. However, it is not clear exactly where the transition occurs, so the datapoints at time **t1** or time **t2** would be discarded by the process **200** (FIG. **2**) before calculating an HVAC effect for a given environment (e.g., at **214**) or determining a calibration value. If it were necessary to determine an altitude using the datapoints at time **t1** or time **t2**, then the uncertainty in the altitude estimate would be relatively large.

[0103] Additionally, the datapoint at time **t0** is the best data in this example for calculating a calibration value or altitude, since it is likely to be outside, so HVAC effects do not have to be taken into consideration for calibration or altitude. On the other hand, the datapoints at times **t3** and **t4** are highly likely to be subject to HVAC effects, so an altitude determination would take this into consideration, but a calibration would preferably not be determined unless it was necessary, e.g., the current calibration value is too stale.

[0104] FIG. **8** also shows an example of five datapoints (**T0-T4** with upper case “T”) for the computing device **102** moving from a freezer environment to an outdoor environment. This example illustrates a path of the computing device **102** as it is transitioning from a cold freezer temperature (potentially subject to HVAC effects) to a warm outdoor temperature (not subject to HVAC effects). This example shows how changes in the device temperature compared with a reference temperature over a given period of time can indicate the location context of the computing device **102**. The direction of the path provides confidence that the computing device **102** starts at an HVAC environment and enters a non-HVAC environment at the later datapoints.

[0105] The datapoint at time **T0** is within the region **824**, so the computing device **102** was likely to have been inside a freezer (e.g., a walk-in freezer) at this time, i.e., the location context is in an HVAC environment. The datapoint at time **T1** is in the region **820**, so the computing device **102** has likely transitioned from the freezer environment to a warmer environment and the device temperature is in the process of transitioning to the warmer temperature during this time. However, it is unknown at time **T1** whether the computing device **102** has entered an indoor HVAC environment or an outdoor non-HVAC environment, so the location context is an uncertain environment at this time. The datapoint **T2** is on the edge between the region **804** and the region **808**, so again the location context is uncertain for this time, since the computing device **102** could be outside or inside. The datapoint at time **T3** is within the region **806**, which is likely outside, so the location context is likely to be in a non-HVAC environment at this time. The datapoint at time **T4** is further within the region **806** but on the edge with region **822**. Due to being on the edge between regions **806** and **822**, the location context is an uncertain environment, or the temperature measurement data could possibly be erroneous, so the process **200** would discard the current datapoint as likely being unreliable.

[0106] The datapoints (**T0** and **T3**) that provide a likely location context, if taken in isolation from each other, would result in a possible uncertainty level or value for the location context determination. However, it is clear that the device temperature is getting warmer (as time passes from **T0** to **T4**), while the reference temperature is staying warm. In this situation, the process **700** determines that the computing device **102** transitioned from a likely HVAC environment (i.e., the freezer at **T0**) through an uncertain environment (**T1** and **T2**) to a likely non-HVAC environment (at **T3**) followed by an unreliable situation (at **T4**). However, the context for this transition provides evidence that the location context determinations for the datapoints **T1**, **T2** and **T3** may be uncertain or unreliable, since the final destination at **T4** may be considered unreliable and the intermediate datapoints **T1**, **T2** and **T3** may all have resulted when the device temperature was in the process of transitioning to the time **T4**. Thus, even though the datapoint at time **T3** might seem in isolation to be a good measurement in a non-HVAC environment which could produce a reliable altitude determination or calibration calculation, the overall context indicates that the datapoint at time **T3** is uncertain or unreliable. Given the overall context of the datapoints from **T0** to **T4**, therefore, if it were necessary to determine an altitude using any of the datapoints, then the

uncertainty in the altitude estimate would be relatively large. Additionally, it would be undesirable to use any of the datapoints for a calibration calculation.

[0107] In some embodiments, although not shown in the flowcharts for the processes **500**, **600** and **700**, each of these processes determines an uncertainty level or confidence level for the result of the location context being in an HVAC environment, a non-HVAC environment, or an uncertain environment.

[0108] The uncertainty or confidence can be defined, for example, as uncertainty or confidence in each of the individual techniques, tests or options described above (e.g., with respect to **504**, **604** and **706/708**) for each of the processes **500**, **600** and **700**. For example, the magnitude of the key ratio between the device footprint and the building footprint (i.e., the proportion of overlap of the device footprint with the building footprint) may be directly proportional to the confidence or indirectly proportional to the uncertainty. As another example, relative weighted confidence or uncertainty values can be empirically determined for each of the techniques, tests or options. As another example, datapoints (e.g., the datapoints at times t_0 and t_4 in FIG. 8) that are within a region of high likelihood for either indoors or outdoors and that are relatively far from an edge of the region would have a lower uncertainty or greater confidence than for those datapoints (e.g., the datapoints at times t_1 - t_3) that are nearer to or are at a region edge.

[0109] The uncertainties or confidences resulting from the individual techniques, tests or options of the processes **500**, **600** and **700** are propagated through the process **200** to contribute to the uncertainty or confidence in the calculated HVAC effect at **214** and then to that of the determined calibration value or estimated altitude at **216**. In other words, a higher uncertainty in the determination of whether the computing device **102** is in an HVAC environment would result in a higher uncertainty in the calculated calibration value or altitude estimation that used the data from these points, and vice versa.

[0110] In some embodiments, a single uncertainty measurement percentage is provided for all environments (i.e., HVAC, non-HVAC, and uncertain environments). In this case, an uncertainty from 0% to about 20% (e.g., $0\% \leq \text{uncertainty} \leq 20\%$) is for a non-HVAC environment. Additionally, an uncertainty from about 80% to 100% (e.g., $80\% \leq \text{uncertainty} \leq 100\%$) is for an HVAC environment. Furthermore, an uncertainty from about 20% to about 80% (e.g., $20\% < \text{uncertainty} < 80\%$) is for an uncertain environment.

[0111] In this disclosure, the processes **500**, **600** and **700** are illustrated and described separately. However, in some embodiments, the processes **500**, **600** and **700** are combined into a single process that performs some or all of the options described herein to determine whether the computing device **102** is in a building, a vehicle, an unspecified HVAC environment, a non-HVAC environment, or an uncertain environment.

[0112] FIG. 10 illustrates components of an example of the reference pressure stations **106**, an example of the computing devices **102**, and an example of the system server **104** of the operational environment **100** shown in FIG. 1, in accordance with some embodiments. Examples of communication pathways are shown by arrows between components. The components shown in FIG. 10 are operable to perform all or a portion of the processes **200**, **300**, **500**, **600** and **700**.

[0113] By way of example in FIG. 10, each of the reference pressure stations **106** may include: a computing device interface **11** for exchanging information with the computing devices **102** (e.g., antenna(s) and RF front-end components known in the art or otherwise disclosed herein); one or more processor(s) **12**; memory/data source **13** for providing storage and retrieval of information and/or program instructions; atmospheric sensor(s) **14** for measuring environmental conditions (e.g., pressure, temperature, humidity, other) at or near the reference pressure station; a server interface **15** for exchanging information with the system server **104** (e.g., an antenna, a network interface, or other); and any other components known to one of ordinary skill in the art. The memory/data source **13** may include a non-transitory memory storing software modules with executable instructions, and the processor(s) **12** may perform different actions by executing the

instructions from the modules, including (i) performance of a part or all of the methods or processes as described herein or otherwise understood by one of skill in the art as being performable at the reference pressure station; (ii) generation of positioning signals for transmission using a selected time, frequency, code, and/or phase; (iii) processing of signals received from the computing device or another source; or (iv) other processing as required by operations described in this disclosure. Signals generated and transmitted by a reference pressure station may carry different information that, once determined by a computing device or a server, may identify the following: the reference pressure station; the reference pressure station's position; environmental conditions at or near the reference pressure station; and/or other information known in the art. The atmospheric sensor(s) **14** may be integral with the reference pressure station, or separate from the reference pressure station and either co-located with the reference pressure station or located in the vicinity of the reference pressure station (e.g., within a threshold amount of distance).

[0114] By way of example in FIG. **10**, the computing device **102** may include: a reference pressure station interface **21** for exchanging information with a reference pressure station (e.g., an antenna and RF front-end components known in the art or otherwise disclosed herein); one or more processor(s) **22**; memory/data source **23** for providing storage and retrieval of information and/or program instructions; atmospheric sensor(s) **24** (such as barometers and temperature sensors) for measuring environmental conditions (e.g., pressure, temperature, other) at the computing device; other sensor(s) **25** for measuring other conditions (e.g., inertial sensors for measuring movement and orientation); a user interface **26** (e.g., display, keyboard, microphone, speaker, other) for permitting a user to provide inputs and receive outputs; a network interface **27** for exchanging information with the system server **104** or other devices external to the computing device (e.g., an antenna, a network interface, or other); and any other components known to one of ordinary skill in the art. A GNSS interface and processing unit (not shown) are contemplated, which may be integrated with other components (e.g., the interface **21** and the processors **22**) or a standalone antenna, RF front end, and processors dedicated to receiving and processing GNSS signaling. The memory/data source **23** may include a non-transitory memory (e.g., a data storage module) storing software modules with executable instructions, and the processor(s) **22** may perform different actions by executing the instructions from the modules, including (i) performance of a part or all of the methods as described herein or otherwise understood by one of ordinary skill in the art as being performable at the computing device; (ii) estimation of an altitude of the computing device based on measurements of pressure from the computing device and reference pressure station(s), temperature measurement(s) from the reference pressure station(s) or another source, and any other information needed for the computation; (iii) processing of received signals to determine position information (e.g., times of arrival or travel time of the signals, pseudoranges between the computing device and reference pressure stations, reference pressure station atmospheric conditions, reference pressure station and/or locations or other reference pressure station information); (iv) use of position information to compute an estimated position of the computing device; (v) determination of movement based on measurements from inertial sensors of the computing device; (vi) GNSS signal processing; (vii) determination of location context; (viii) determination of HVAC effect; or (ix) other processing as required by operations described in this disclosure.

[0115] By way of example in FIG. **10**, the system server **104** may include: a computing device interface **31** for exchanging information with a computing device (e.g., an antenna, a network interface, or other); one or more processor(s) **32**; memory/data source **33** for providing storage and retrieval of information and/or program instructions; a reference pressure station interface **34** for exchanging information with a reference pressure station (e.g., an antenna, a network interface, or other); and any other components known to one of ordinary skill in the art. The memory/data source **33** may include a non-transitory memory storing software modules with executable instructions, and the processor(s) **32** may perform different actions by executing instructions from

the modules, including (i) performance of a part or all of the methods as described herein or otherwise understood by one of ordinary skill in the art as being performable at the server; (ii) estimation of an altitude of the computing device; (iii) computation of an estimated position of the computing device; (iv) determination of location context; (v) determination of HVAC effect; or (vi) other processing as required by operations described in this disclosure. Steps performed by servers as described herein may also be performed on other machines that are remote from a computing device, including computers of enterprises or any other suitable machine.

[0116] Certain aspects disclosed herein relate to estimating the positions of computing devices—e.g., where the position is represented in terms of latitude, longitude, and/or altitude coordinates; x, y, and/or z coordinates; angular coordinates; or other representations. Various techniques to estimate the position of a computing device can be used, including trilateration, which is the process of using geometry to estimate the position of a computing device using distances traveled by different “positioning” (or “ranging”) signals that are received by the computing device from different beacons (e.g., terrestrial transmitters and/or satellites). If position information like the transmission time and reception time of a positioning signal from a beacon is known, then the difference between those times multiplied by the speed of light would provide an estimate of the distance traveled by that positioning signal from that beacon to the computing device. Different estimated distances corresponding to different positioning signals from different beacons can be used along with position information like the locations of those beacons to estimate the position of the computing device. Positioning systems and methods that estimate a position of a computing device (in terms of latitude, longitude, and/or altitude) based on positioning signals from beacons (e.g., transmitters, and/or satellites) and/or atmospheric measurements are described in co-assigned U.S. Pat. No. 8,130,141, issued Mar. 6, 2012, and U.S. Pat. No. 9,057,606, issued Jun. 16, 2015, incorporated by reference herein in their entirety for all purposes. It is noted that the term “positioning system” may refer to satellite systems (e.g., Global Navigation Satellite Systems (GNSS) like GPS, GLONASS, Galileo, and Compass/Beidou), terrestrial transmitter systems, and hybrid satellite/terrestrial systems.

[0117] Reference has been made in detail to embodiments of the disclosed invention, one or more examples of which have been illustrated in the accompanying figures. Each example has been provided by way of an explanation of the present technology, not as a limitation of the present technology. In fact, while the specification has been described in detail with respect to specific embodiments of the invention, it will be appreciated that those skilled in the art, upon attaining an understanding of the foregoing, may readily conceive of alterations to, variations of, and equivalents to these embodiments. For instance, features illustrated or described as part of one embodiment may be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present subject matter covers all such modifications and variations within the scope of the appended claims and their equivalents. These and other modifications and variations to the present invention may be practiced by those of ordinary skill in the art, without departing from the scope of the present invention, which is more particularly set forth in the appended claims. Furthermore, those of ordinary skill in the art will appreciate that the foregoing description is by way of example only, and is not intended to limit the invention.

Claims

1. A method comprising: determining a location of a computing device; determining a location context for the computing device, wherein the location context indicates whether the computing device is in an HVAC (heating, ventilating, or air-conditioning) environment that can be subject to an HVAC effect or is in a non-HVAC environment that is not subject to the HVAC effect; upon determining that the computing device is in the non-HVAC environment, not determining the HVAC effect; and upon determining that the computing device is in the HVAC environment:

obtaining an expected altitude of the computing device based on the location of the computing device; determining a measured altitude of the computing device based on a reference pressure and a measured pressure from a pressure sensor of the computing device; determining an altitude difference between the expected altitude and the measured altitude; determining the HVAC effect based on the altitude difference; and calibrating the pressure sensor based on the HVAC effect or estimating a corrected measured altitude of the computing device based on the HVAC effect.

2. The method of claim 1, wherein determining the location context for the computing device further comprises: determining a device temperature of the computing device; obtaining a reference temperature based on the location of the computing device; and determining the location context by comparing the device temperature and the reference temperature.

3. The method of claim 2, wherein determining the location context for the computing device further comprises: determining that the reference temperature and the device temperature are within a threshold temperature range from each other; determining that the reference temperature and the device temperature are both less than a cold threshold temperature; and determining that the computing device is in the non-HVAC environment.

4. The method of claim 2, wherein determining the location context for the computing device further comprises: determining that the reference temperature and the device temperature are within a threshold temperature range from each other; determining that the reference temperature and the device temperature are both greater than a hot threshold temperature; and determining that the computing device is in the non-HVAC environment.

5. The method of claim 2, wherein: the location context further indicates whether the computing device is in the HVAC environment, the non-HVAC environment, or an uncertain environment; and determining the location context for the computing device further comprises: determining that the reference temperature and the device temperature are within a threshold temperature range from each other; determining that the reference temperature and the device temperature are both greater than a cold threshold temperature and less than a hot threshold temperature; and determining that the computing device is in the uncertain environment.

6. The method of claim 2, wherein determining the location context for the computing device further comprises: determining that the reference temperature is warmer than the device temperature; determining that the device temperature is greater than a cold threshold temperature and less than a hot threshold temperature; and determining that the computing device is in the HVAC environment.

7. The method of claim 2, wherein determining the location context for the computing device further comprises: determining that the reference temperature is colder than the device temperature; determining that the device temperature is greater than a cold threshold temperature and less than a hot threshold temperature; and determining that the computing device is in the HVAC environment.

8. The method of claim 2, wherein: the location context further indicates whether the computing device is in the HVAC environment, the non-HVAC environment, or an uncertain environment; and determining the location context for the computing device further comprises: determining that the reference temperature is colder than the device temperature; determining that the device temperature is near a cold threshold temperature within a low-end temperature deviation amount; and determining that the computing device is in the HVAC environment or the uncertain environment.

9. The method of claim 2, wherein: the location context further indicates whether the computing device is in the HVAC environment, the non-HVAC environment, or an uncertain environment; and determining the location context for the computing device further comprises: determining that the reference temperature is warmer than the device temperature; determining that the device temperature is near a hot threshold temperature within a high-end temperature deviation amount; and determining that the computing device is in the HVAC environment or the uncertain environment.

- 10.** The method of claim 2, wherein determining the location context for the computing device further comprises: determining that the reference temperature is colder than the device temperature; determining that the device temperature is below a cold threshold temperature; and determining that the computing device is in the non-HVAC environment.
- 11.** The method of claim 2, wherein: the location context further indicates whether the computing device is in the HVAC environment, the non-HVAC environment, or an uncertain environment; and determining the location context for the computing device further comprises: determining that the reference temperature is warmer than the device temperature; determining that the device temperature is above a hot threshold temperature; and determining that the computing device is in the uncertain environment.
- 12.** The method of claim 2, wherein determining the location context for the computing device further comprises: determining the location context by comparing the device temperature and the reference temperature and further determining how the device temperature changes over time.
- 13.** The method of claim 1, wherein: the steps from the determining of the location of the computing device to the determining of the altitude difference are repeated for each time point of a plurality of time points; and the determining of the HVAC effect further comprises: for each time point of the plurality of time points, mapping the altitude difference with respect to the location context to generate mapped altitude differences; determining a first cluster of the mapped altitude differences for when the computing device is in the HVAC environment; determining a second cluster of the mapped altitude differences for when the computing device is in the non-HVAC environment; and determining the HVAC effect based on the first cluster of the mapped altitude differences and the second cluster of the mapped altitude differences.
- 14.** The method of claim 13, wherein the determining of the HVAC effect further comprises: for each time point of the plurality of time points, mapping the altitude difference with respect to the time point and the location context to generate the mapped altitude differences.
- 15.** The method of claim 13, wherein the determining of the HVAC effect further comprises: determining a magnitude of the HVAC effect that is a difference between a first average of the first cluster of the mapped altitude differences and a second average of the second cluster of the mapped altitude differences.
- 16.** The method of claim 1, wherein determining the location context for the computing device further comprises: determining that the location context is in the HVAC environment which is a building.
- 17.** The method of claim 16, wherein determining the location context for the computing device further comprises: determining that the computing device is in the building by mapping the location of the computing device to a building database, which indicates that the computing device is within the building; determining that the computing device is in the building by determining that the computing device is wirelessly connected to a wireless access point that is known to be in the building, which indicates that the computing device is within the building; or determining that the computing device is in the building by determining that a strength of a satellite signal is attenuated from a level that it should be if the computing device were outdoors, which indicates that the computing device is within the building.
- 18.** The method of claim 1, wherein determining the location context for the computing device further comprises: determining that the location context is in the HVAC environment which is a vehicle.
- 19.** The method of claim 18, wherein determining the location context for the computing device further comprises: determining that the computing device is in the vehicle by mapping the location of the computing device to a road database, which indicates that the computing device is on a road and within the vehicle; determining that the computing device is in the vehicle by determining that the computing device is communicatively connected to the vehicle, which indicates that the computing device is within the vehicle; determining that the computing device is in the vehicle by

determining that the computing device is moving faster than a walking, running or biking speed, which indicates that the computing device is within the vehicle; determining that the computing device is in the vehicle by determining that the computing device is changing cell tower connections, which indicates that the computing device is within the vehicle; determining that the computing device is in the vehicle by obtaining an activity context from a sensor hub of the computing device that indicates that the activity context is driving, which indicates that the computing device is within the vehicle; determining that the computing device is in the vehicle by determining that the computing device is in a parking lot, which indicates that the computing device is within the vehicle; or determining that the computing device is not in the vehicle by determining that the computing device is in the parking lot and is moving at a walking speed, does not follow vehicle lanes of the parking lot, or moves in a straight line.

20. A method comprising: determining a location of a computing device; determining a location context for the computing device, wherein the location context indicates whether the computing device is in an HVAC (heating, ventilating, or air-conditioning) environment that can be subject to an HVAC effect or is in a non-HVAC environment that is not subject to the HVAC effect; upon determining that the computing device is in the non-HVAC environment, not determining the HVAC effect; and upon determining that the computing device is in the HVAC environment: determining a first measured pressure from the computing device using a pressure sensor of the computing device when the computing device is in the non-HVAC environment and a second measured pressure from the computing device using the pressure sensor when the computing device is in the HVAC environment; determining a pressure difference between the first measured pressure and the second measured pressure; determining the HVAC effect based on the pressure difference; and calibrating the pressure sensor based on the HVAC effect or estimating a measured altitude of the computing device based on the HVAC effect.
